# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 6 — MATHS · MEDIUM LEVEL UNIT TEST SET 1

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** A rectangle has perimeter 50 cm and length 18 cm. Its area (in $cm^2$) is: 1

**(a)** 126 **(b)** 144 **(c)** 108 **(d)** 133

**2.** $5\frac{1}{2} - 2\frac{1}{3} =$ 1

**(a)** $3\frac{1}{2}$ **(b)** $4$ **(c)** $3\frac{5}{6}$ **(d)** $3\frac{1}{6}$

**3.** Two circles have radii 8 cm and 5 cm and their centres are 10 cm apart. The circles: 1

**(a)** do not touch (separate) **(b)** touch internally

**(c)** intersect at two points **(d)** touch externally

**4.** A regular polygon has each angle of rotation $20°$. The number of lines of symmetry is: 1

**(a)** 18 **(b)** 20 **(c)** 17 **(d)** 19

**5.** -15 + 8 + (-6) = 1

**(a)** $-13$ **(b)** $-11$ **(c)** $-15$ **(d)** $-3$

**6.** The area of a square is 36 $cm^2$. If its side is doubled, the new area (in $cm^2$) is: 1

**(a)** 144 **(b)** 156 **(c)** 150 **(d)** 138

**7.** $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} =$ 1

**(a)** $\frac{5}{6}$ **(b)** $\frac{3}{11}$ **(c)** $1$ **(d)** $\frac{2}{3}$

**8.** A square has a diagonal that splits it into two triangles. If the side is 5 cm, the sum of the angles of one triangle is (degrees): 1

**(a)** 540 **(b)** 90 **(c)** 180 **(d)** 270

**9.** The total angle through which a regular hexagon turns when rotated 5 times through its smallest angle (degrees): 1

**(a)** 310 **(b)** 300 **(c)** 330 **(d)** 290

**10.** 10 - (-4) - 6 = 1

**(a)** 8 **(b)** 10 **(c)** 18 **(d)** 6

**11.** A rectangular room 8 m by 5 m has a carpet costing ₹120 per $m^2$. The cost of carpeting (₹): 1

**(a)** 4900 **(b)** 4680 **(c)** 4800 **(d)** 4920

**12.** $\frac{3}{5}$ of a number is 36. The number is: 1

**(a)** 48 **(b)** 72 **(c)** 60 **(d)** 63

**13.** A rectangle's diagonals meet at O making $\angle AOB = 50^\circ$. Then $\angle AOD$ is (degrees): 1

**(a)** 160 **(b)** 130 **(c)** 120 **(d)** 140

**14.** A figure has 4 lines of symmetry. Which could it be? 1

**(a)** A rectangle **(b)** A square

**(c)** A scalene triangle **(d)** A rhombus (not a square)

**15.** The sum of all integers from -4 to 4 is: 1

**(a)** $0$ **(b)** $4$ **(c)** $1$ **(d)** $-1$

## SECTION — B (Very Short Answer Type) 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** The perimeter of a rectangle is 70 cm and its length is 24 cm. Find the breadth and the area. 2

**17.** Find $5\frac{3}{4}$ - $2\frac{1}{2}$. 2

**18.** Two circles with radii 9 cm and 4 cm touch externally. Find the distance between their centres, and if they touch internally. 2

**19.** A regular polygon rotates through $40^\circ$ to coincide with itself. How many sides does it have? Name its order of symmetry. 2

**20.** Find $-18 + 25 + (-9)$. 2

## SECTION — C (Short Answer Type) 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** A rectangular lawn 30 m by 20 m has a path 2 m wide inside it all around. Find the area of the path. 3

**22.** A farmer ploughed $\frac{1}{4}$ of a field on Monday and $\frac{1}{3}$ of it on Tuesday. What part of the field remains to be ploughed? 3

**23.** A square of side 6 cm has its diagonals drawn. Find the area of one of the four triangles formed (it is a quarter of the square) and the sum of angles at the centre. 3

**24.** A regular polygon has 5 lines of symmetry. Find its angle of rotation, the angle after 3 turns and the number of turns to make a full rotation. 3

OR

A train station has platforms numbered with 0 at ground level. A lift moves 3 floors below ground, then 5 floors up, then 4 floors below again. Find its final floor and the total number of floors moved. 3

**25.** Two rectangles have the same perimeter 48 cm: one is 20 cm by 4 cm and the other is 12 cm by 12 cm. Which has the larger area and by how much? 3

OR

Find the value of $2\frac{1}{2}$ + $\frac{1}{3}$ - $\frac{1}{4}$. 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** A rectangular hall 10 m by 8 m is to be covered with square tiles of side 50 cm costing ₹40 each. Find the number of tiles and the total cost. 5

OR

Ravi had ₹120. He spent $\frac{1}{4}$ of it on books and $\frac{1}{3}$ of the rest on a bag. How much does he have left? 5

**27.** Write the steps to construct a square ABCD of side 4 cm using only ruler and compass. After constructing, state the lengths of both diagonals' halves if the full diagonal is 6 cm and find the angle formed by the diagonals. 5

**OR**

A rangoli is a regular 5-pointed pattern. Find 5

(a) its lines of symmetry,

(b) the smallest rotation angle,

(c) how many rotations of that angle bring it back to the start,

(d) whether a rotation of $144^\circ$ leaves it unchanged.

**END OF QUESTION PAPER** **Total: 50 Marks**

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